

Constraint-Based Problem Solving

Computer Networking Solutions & Analysis

1. Startup Network Architecture Proposal

Network Topology:

- **Star Topology (Extended Star)** on each floor, connected via a single central backbone link.

Justification: Simple to troubleshoot, isolate faults, and add new workstations without disturbing the rest of the floor network.

Transmission Media:

- **Intra-Floor (Workstations to Switch):** Cat6 Ethernet Cabling — Cost-effective, supports up to 1 Gbps (and 10 Gbps for short distances), and simple to route within open floor layouts.
- **Inter-Floor (Floor 1 to Floor 2):** Single Cat6a or OM3 Fiber Optic link — Requires only a single cabling run between floors (respecting physical installation limits) while ensuring high-bandwidth trunking.

Devices:

- **1x Dual-WAN Router / Firewall:** Located on Floor 1 for network edge protection, security routing, and ISP failover redundancy.
- **2x 28-Port Managed Gigabit Ethernet Switches (with PoE):** One switch per floor to connect 25 users alongside extra capacity for printers and Access Points.
- **2–4x Wi-Fi 6 Access Points:** Handles wireless guest access and mobile devices without burning physical switch ports.

Constraint Balancing:

- **Cost:** Managed Gigabit switches and Cat6 cabling offer enterprise reliability without the high expense of complex mesh hardware.
- **Scalability:** Unused switch ports (56 total ports for 50 users) and expandable Wi-Fi permit seamless onboarding of new employees.
- **Cabling Limits:** A single high-speed backbone link between floors eliminates inter-floor conduit complexity and wall drilling.

2. Educational Institution Traffic Management & QoS Strategy

Traffic Management & Quality of Service Techniques:

- **Bandwidth Throttling & Rate Limiting:** Enforce strict per-user speed caps on non-essential traffic (e.g., maximum download speed caps for general browsing and P2P/BitTorrent protocols).

- **Application Filtering & Content Blocking:** Temporarily block or restrict high-bandwidth, non-educational video streaming services (e.g., Netflix, YouTube 4K, social media apps) during core instruction hours.
- **Quality of Service (QoS) Prioritization Queues:**
 - **Priority 1 (High / Strict Priority):** Real-time interactive traffic including online live classes, Zoom/Teams web conferencing, and VoIP calling.
 - **Priority 2 (Medium Priority):** Essential academic systems, LMS platforms (Canvas, Moodle), and administrative database access.
 - **Priority 3 (Low / Best Effort):** General web browsing, video streaming services, and background software updates.
- **Traffic Shaping & Caching:** Deploy local **Proxy Caching Servers** to cache frequently accessed educational media and documents locally, reducing repetitive external WAN traffic.

Constraint Resolution:

- These software-driven, logical traffic management policies optimize existing throughput immediately, ensuring critical interactive learning operates smoothly without requiring capital expenditure on upgraded infrastructure.

3. Hospital High-Reliability Network Design

Network Topology Selection:

- **Partial Mesh Topology combined with Spine-and-Leaf Architecture:**

Justification: A full mesh network requires $N(N - 1) / 2$ links, making it cost-prohibitive. Partial mesh connects only core critical nodes (Emergency Room, ICU, Server Room) redundantly, dramatically cutting hardware/cabling costs while maintaining continuous uptime for life-critical systems.

Redundancy Techniques:

- **Hardware/Core Redundancy:** Dual Core Switches operating in a high-availability active-passive or active-active cluster using **VRRP** (Virtual Router Redundancy Protocol) or **HSRP**.
- **Link Aggregation & Path Redundancy:** **LACP (EtherChannel)** for bundling physical links between core switches, alongside **RSTP** (Rapid Spanning Tree Protocol) or **MSTP** to avoid loops while providing automatic sub-second failover path routing if a cable is cut.
- **Dual-Homing:** Direct connection of vital patient monitoring units and critical department switches via redundant links to both core switches.
- **Uninterruptible Power Supply (UPS):** Dedicated battery backups and redundant dual power supplies (PoE switches) to ensure monitoring nodes endure localized power disruptions.

Constraint Balancing:

- Focuses redundancy budget specifically on life-sustaining infrastructure (partial mesh) while utilizing standard dual-homed topologies for non-critical admin zones, delivering zero-downtime reliability within budget limits.